ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2

PAPER 2 November 2021

2 hours

PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2021 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [ ].
INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

1. (a) Name two structures found in a plant cell but not in an animal cell. [2]<<
>>(b) State the function of the cell membrane. [2]<<
>>(c) Describe how to prepare a slide of onion epidermis to view under a light microscope. [3]<<
>>(d) Explain why muscle cells contain many mitochondria. [3] **[10]**

2. (a) Write the word equation for aerobic respiration. [2]<<
>>(b) Name the organelle where aerobic respiration occurs. [1]<<
>>(c) Distinguish between aerobic and anaerobic respiration in human muscle. [4]<<
>>(d) Suggest why a sprinter may have an oxygen debt after a race. [3] **[10]**

3. (a) State what is transported in xylem and in phloem. [2]<<
>>(b) Define transpiration. [2]<<
>>(c) Explain how the structure of xylem is adapted to its function. [3]<<
>>(d) Suggest two environmental conditions that increase the rate of transpiration. [3] **[10]**

4. (a) State where sperm are produced and where fertilisation occurs in humans. [2]<<
>>(b) Name two male secondary sexual characteristics. [2]<<
>>(c) Describe one barrier method of contraception and how it works. [3]<<
>>(d) Explain how the placenta is important to the fetus. [3] **[10]**

5. (a) Describe how an ionic bond forms between sodium and chlorine. [3]<<
>>(b) Describe a covalent bond. [2]<<
>>(c) Explain why sodium chloride has a high melting point. [3]<<
>>(d) Explain why chlorine gas, Cl_2, has a low melting point. [2] **[10]**

6. (a) What is an electrolyte? [2]<<
>>(b) Name the products at the cathode and anode when molten lead(II) bromide is electrolysed. [2]<<
>>(c) Explain why solid lead(II) bromide does not conduct electricity but molten lead(II) bromide does. [3]<<
>>(d) State one industrial use of electrolysis. [3] **[10]**

7. (a) Define average speed. [2]<<
>>(b) A learner runs 100 m in 5 s. Calculate the average speed. [3]<<
>>(c) A car travels at constant velocity. What is the resultant force on the car? Explain. [3]<<
>>(d) Name the instrument used to measure time in this experiment. [2] **[10]**

8. (a) State the law of reflection. [2]<<
>>(b) A ray in air hits a glass block. Describe what happens to its speed and direction. [3]<<
>>(c) Explain why sound cannot travel through a vacuum, but light can. [3]<<
>>(d) State how pitch and loudness of a sound relate to the wave. [2] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).